

Lesson 01 Practice: Square and Cube Roots

Name: _____

Class: MYP 8A

Date: _____

How to use this sheet

Estimated time: 25 minutes. **Required:** R1, Easy (H1–H6), Medium (H7–H12) and the Self-Test. **Optional challenge:** Hard (H13–H16).

Short on time? Do R1, H1, H2, H4, H5, H7, H10, H13 and the Self-Test.

Calculator: you may use it to multiply (e.g. 6.4×6.4), but do *not* use the $\sqrt{\quad}$ or $\sqrt[3]{\quad}$ keys. Show your working.

Remember: the five steps for estimating

Trap between perfect squares/cubes → **Roots** (two whole numbers) → **Closer?** → **Test** decimals → **Decide** ($\approx$).

Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225 Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

Retrieval

from earlier in Unit 1

R1. Evaluate (a) 10^3 (b) $(-2)^4$

Easy: Build confidence

H1–H6

H1. (a) $\sqrt{36} = \underline{\quad}$ (b) $\sqrt{81} = \underline{\quad}$ (c) $\sqrt{144} = \underline{\quad}$

H2. (a) $\sqrt[3]{27} = \underline{\quad}$ (b) $\sqrt[3]{1000} = \underline{\quad}$ (c) $\sqrt[3]{-8} = \underline{\quad}$

H3. Complete: $\sqrt{169} = \underline{\quad}$ because $\underline{\quad}^2 = 169$

H4. Between which two consecutive whole numbers does each root lie?

(a) $\underline{\quad} < \sqrt{12} < \underline{\quad}$ (b) $\underline{\quad} < \sqrt{70} < \underline{\quad}$ (c) $\underline{\quad} < \sqrt[3]{10} < \underline{\quad}$

H5. A square has area 64 cm^2 . Find the length of its side. **H6.** A cube has volume 216 cm^3 . Find the length of its edge.

Medium: Show understanding

H7–H12

H7. Estimate $\sqrt{42}$ to 1 decimal place. Show your trap and your test.

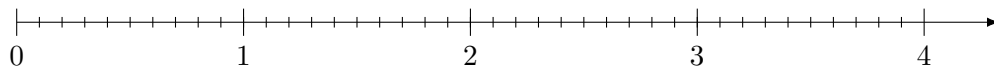
H8. Estimate $\sqrt[3]{60}$ to 1 decimal place. Show your trap and your test.

H9. Which is larger, $\sqrt{26}$ or $\sqrt[3]{100}$? Explain *without* working out either root exactly.

H10. Explain why $\sqrt[3]{-64}$ is a real number but $\sqrt{-64}$ is not.

H11. Mark these on the number line, then list them from smallest to largest.

$\sqrt{5}$, $\sqrt{15}$, $\sqrt[3]{9}$, $\sqrt[3]{27}$, 3.5



Order: _____

H12. Fill in the missing numbers.

(a) $\sqrt{\boxed{}} = 20$

(b) $\sqrt[3]{\boxed{}} = -3$

(c) $\sqrt{\boxed{}} = 0.5$

Hard: Apply and reason***H13–H16 · optional if short on time***

H13. A square rug has area 30 m^2 . Will it fit flat across a hallway that is 5.2 m wide? Justify your answer.

H14. Kai estimated $\sqrt[3]{70}$ like this:

Kai's work

$64 < 70 < 125$, so $4 < \sqrt[3]{70} < 5$. 70 is between 64 and 125, so $\sqrt[3]{70} \approx 4.5$.

(a) What is Kai's first mistake? (b) Find a better estimate to 1 decimal place and check it.

H15. Find **all** the whole numbers n for which $\sqrt{n}$ is between 6 and 7. How many are there? Explain how you know you have found them all.

H16. A cube has the same volume as a cuboid measuring $4 \text{ cm} \times 4 \text{ cm} \times 8 \text{ cm}$. Estimate the edge length of the cube to 1 decimal place.

Self-Test*about 3 minutes · no calculator*

ST1. $\sqrt{121} =$

ST2. $\sqrt[3]{-216} =$

ST3. Estimate $\sqrt{55}$ to 1 d.p.

Reflection**I can do independently:**

I need to practise:

My question:
